Training Day 20

1. Topics Covered and Brief Explanation

1. Image Resizing: Resizing changes the dimensions of an image. It is useful when images need to be adjusted to a fixed size before processing.
2. Image Cropping: Cropping is used to extract a specific region or portion of an image.
3. Colour Conversion: OpenCV allows images to be converted from one colour space to another, such as BGR to grayscale.
4. Image Blurring: Blurring reduces unnecessary details and noise in an image. Gaussian Blur is a commonly used technique.
5. Image Filtering: Filters are applied to images to improve their quality or highlight particular features.
6. Edge Detection: Edge detection identifies boundaries of objects in an image. The Canny Edge Detection method is commonly used in computer vision.

2. Tools and Technologies Used

Tool/Technology	Purpose
Python	Programming language
OpenCV	Image processing operations
NumPy	Handling image arrays
VS Code / Google Colab	Code execution

3. Practical Learning / Code Examples

A basic program was used to apply different image processing techniques.

```
import cv2
image = cv2.imread("image.jpg")
if image is not None
    # Resize image
    resized = cv2.resize(image, (300, 300))
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    # Apply Gaussian blur
    blurred = cv2.GaussianBlur(image, (5, 5), 0)
    # Detect edges
```

```
    edges = cv2.Canny(gray, 100, 200)
    # Save processed images
    cv2.imwrite("resized.jpg", resized)
    cv2.imwrite("gray.jpg", gray)
    cv2.imwrite("blurred.jpg", blurred)
    cv2.imwrite("edges.jpg", edges)

    print("Image processing completed.")

else:
    print("Image not found")
```

Explanation:

- cv2.resize() changes the image dimensions.
- cv2.cvtColor() converts the image into grayscale.
- cv2.GaussianBlur() reduces noise and smooths the image.
- cv2.Canny() detects edges in the image.
- cv2.imwrite() saves the processed images.

4. Learning Outcomes

- Learned basic image processing techniques using OpenCV.
- Understood image resizing and cropping.
- Practised converting colour images into grayscale.
- Learned how blurring helps reduce image noise.
- Understood the basic working of edge detection.